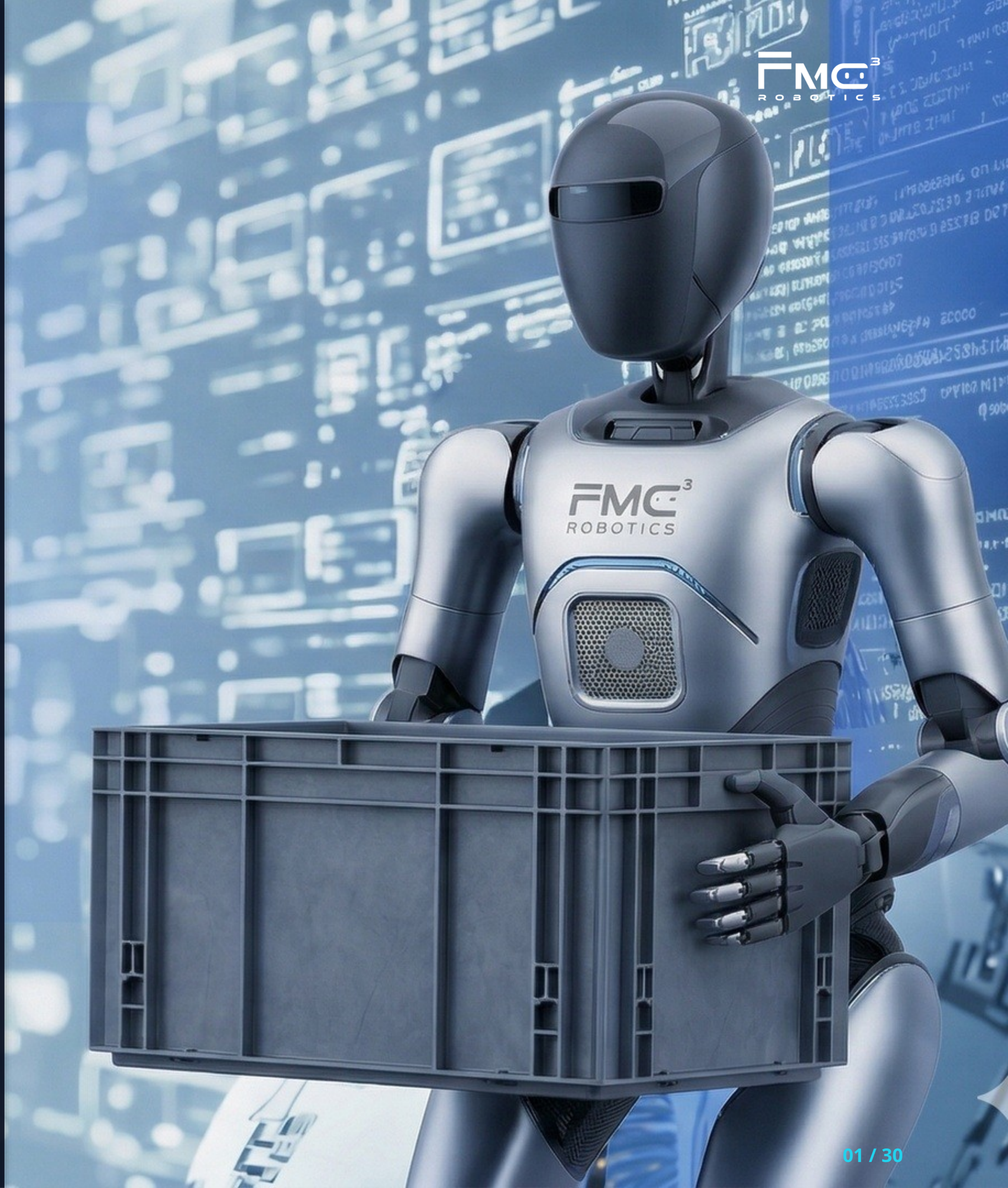
FMC³ Robotics

The Embodied AI Brain from Europe



Product Manual

Mar, 2026
Confidential

Positioning

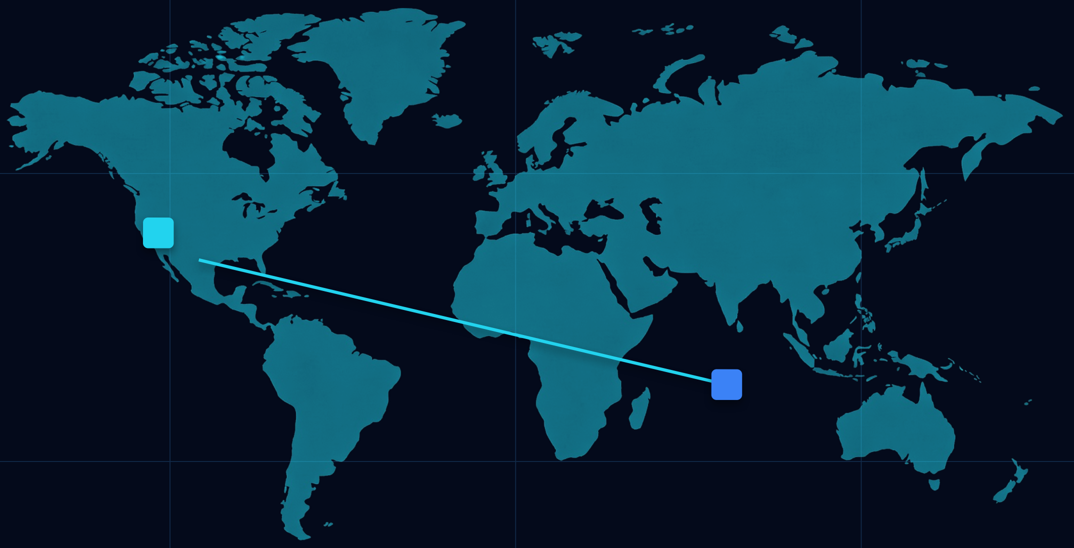
- Maximizing the commercial value of Embodied AI in developed Western markets by leveraging German commercial scenarios and Chinese supply chain excellence.
- Establishing crossscenario standards for Embodied AI through a "Physical AI + ForceTouch Closed Loop" to achieve fullstack autonomy.

Europe HQ

Strategic HQ: Direct market access with a focus on realworld deployment.
Data Moat: Compliance with EU AI Acts and GDPR data regulations.
Training & Delivery: Building the data training closed loop.
Schloßlände 26, Ingolstadt
Bavaria, Germany

China Office

R&D & Design: Development of highfidelity data collection systems.
Hardware Supply Chain: Comprehensive upstream and downstream robotics resources.
Pilot & Testing: Rapid POC/MVP iterations and closedloop training tests.
Jia Xing & Shanghai
China



HQ
Product &Engineering

China office
RD&Product design

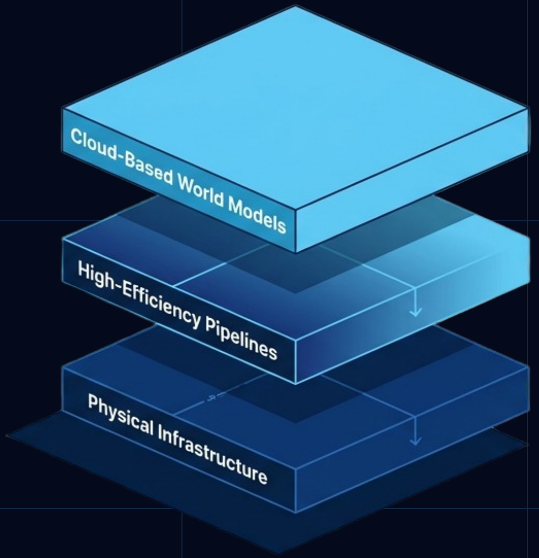
One Brain filt all

Data loop, embodied intelligence and multi-body adaptation

Big Model

Edge Brain

Multi-body adaptation



Humanoid

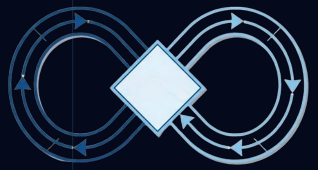
Complex Flexible Assembly

Wheeled Robot

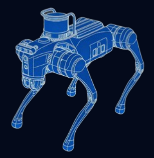
Wide-range Mobile Manipulation

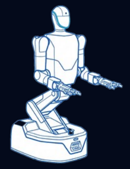
Robotic Arm

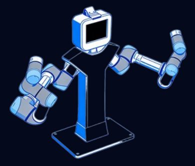
High-precision Station Operation

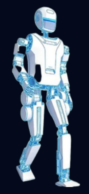


- Long-horizon Task Orchestration
- Zero-code Task Orchestration
- Industry Knowledge Base LLM
- Multi-agent Collaborative Scheduling
- Data
- Loop
- Data Acquisition
- Data Training
- Inference Optimization









Core team

A Team of European Intelligent Systems Scientists

Dr. Wang Cheng

CEO

•Dr.-Ing. from Technical University of Munich (TUM).
•Expert in Autonomous Driving.
•Architect of Audi's Intelligent Driving System.
•Proven track record in the mass production and delivery of over 1 million intelligent driving vehicles.

Prof. Dr. Giesler Bjoern

CTO

•PhD in Computer Science from KIT; a leading authority in AI architecture.
•Holds 50+ core patents in sensing and data training.
•Extensive experience managing international R&D teams of 500+ members.

Fan David, PhD

COO

•Former President of Marelli China.
•Managed business operations with annual revenues exceeding $1.5 billion.
•Led multiple post-M&A organizational integrations and supply chain restructurings.

Dr. Huang Dong

CPO

•Dr.-Ing. in Computer Science from KIT.
•Former Scientist at Siemens Labs and CTO of TerraIT.
•15+ years of expertise in data simulation and Sim-to-Real domains.

Obermeier Florian

MD Germany

•Co-founder of SANEON GmbH.
•16+ years of B2B experience with OEMs and Tier-1 automotive suppliers.
•Global track record serving clients across Europe, Asia, and the US.

GDPR Data Compliance

Localized Data Closed-Loop Architecture (Germany/Europe)

Local Data Loop

Compliant with the EU AI Act

Fully compliant with regulatory requirements for high-risk AI systems, featuring transparent and interpretable auditing capabilities.

Local Collection

Zero Cross-border Transfer. All industrial data stored 100% within German sovereign data centers.

Encrypted

Privacy Computing

Auto-Anonymization. Real-time de-identification of facial & environmental metadata; GDPR compliant.

Anonymized

Federated Learning

Gradient-only Sync. Local data remains on-premise; only model parameters are transmitted.

Secure ML

Principle of data minimization

right of user to be forgotten

storage limitation principle

Limitation Principle of Purpose

Entering the European Industry: Building a Moat Based on "Deterministic, Safe, and Compliant" Standards

Collaborative Safety & Compliance

Machinery Regulation (EU) Core Operational Safety
RED: Wireless Communication Security
EMC Directive
Device Interference-free Operation
EU AI Act
AI Algorithm Governance

1. Safety of Control Systems

Resilience against operational stress and interference.
Must not lead to hazardous situations

1

2.Special Requirements for Autonomous Mobility

Sensing & Stopping: Automatic stop upon detection of humans.
Motion Stability: Anti-tip and anti-loss-of-control measures.

2

3.Risk Monitoring & Cybersecurity

Defense against malicious network manipulation.

3

4. Software Safety & Self-learning

Learning Boundaries: Implementation of strict learning constraints.
OTA Assessment: Mandatory safety re-evaluation for all remote software updates.

4

The Four Hard Requirements for Embodied AI Robots under the "Machinery Regulation"

Our Partners and Clients

- Technical University of Munich (TUM)
- NMI Naturwissenschaftliches und Medizinisches Institut
- KIT Karlsruhe Institute of Technology
- Technische Hochschule Ingolstadt
- HIKAUTO
- INOVANCE
- AGIBOT
- FOURIER
- BMW
- KAUTEX A Textron Company
- FAU Friedrich-Alexander-Universität Erlangen-Nürnberg

FMC^3 Brain+ODM

Robo dog

Commercial & Industrial Cleaning

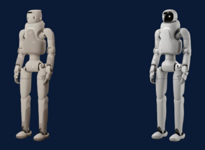
FMC^3 Brain + FMC^3 Data Acquisition Suit

Workbot

Cobot

Industrial Service Robot

Humanoid Home Service Robot

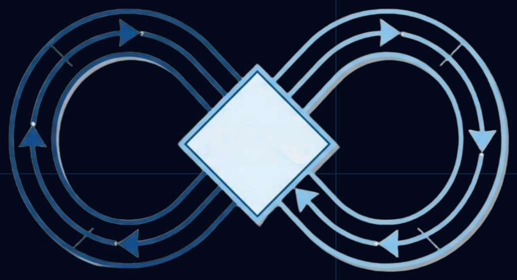


GreetingBot

* All hardware, software, solutions, and related services can be provided either as standalone component or as fully integrated turnkey packages, depending on customer requirement.

FMC³ Brain+Data Acquisition Suit

Main Computing Chips:	**Nvidia Orin / Thor**
Computing Power:	**275 TOPS (Scalable)**
Interface Standards:	**EtherCAT / CAN FD**
Power Consumption:	**15W 60W (Dynamic)**



- FMC³ Brain
- Long-horizon Task Planner
- Code Generator
- LLM Database
- Multi-agent Scheduling
- Data Collection
- Model Training
- Inference

Realtime Teleoperation & Emergency Takeover

LowLatency Streaming: Supports crossborder remote operations.
Dataintheloop: Realtime cleaning and ingestion of teleoperation data into the database.
Safety Critical: Industrialgrade safety takeover mechanism.

FullLink Data Toolchain

Workflow: Collection → Autolabeling → Training Closed Loop.
Compatibility : Fully compatible with NVIDIA Omniverse and Isaac Sim.
Heterogeneous Data Fusion: Seamless integration of Realworld + Simulation data.

Core Data Set

Industrial Haptic MultiAgent Dataset
Proprietary industrialgrade dataset featuring haptic/force modalities.
Covers 3,000+ core industrial processes, including precision assembly, flexible sorting, and surface polishing.
Joint R&D: Shanghai Jiao Tong University (SJTU) Robotics Institute

FMC^3 Omni-Grab : Ultra-Fidelity, Body-Free Human-Ego-Centric Data Capture System

Active IR Stereo RGB-D Headset

Delivers first-person 3D depth spatial perception. Both depth and color run at 1280×720 @ 30fps with an integrated 6-axis IMU (50Hz) for dynamic tracking.

First-person geometric and semantic depth alignment.

400-Point High-Density Haptic Glove

32 sensors per finger, trigger force as low as 30g, max range 1500g, with 200Hz haptic stream high-speed export.

Delivers the force-feedback ground truth most critical for imitation learning

Portable Data Sync Hub

Supports 16 hours of continuous single-session recording

Handles global millisecond-grade hardware time alignment, local caching, and data security.

End-to-End Automated Processing Pipeline

DATA REFINERY PIPELINE

STEP 1 — **High-Precision Synchronized Capture:**

Raw video, IMU, haptic, and EMG streams are hardware-stamped and losslessly aligned to disk.

STEP 2 — **Automatic Scene Segmentation & Slicing:**

ORB-SLAM3-powered segmentation automatically slices semantically rich task episodes by action and odometry.

STEP 3 — **Smart Cleaning & Multi-Dimensional QA:**

Physics-plausibility validation automatically filters corrupted frames, body penetrations, and violent jitter.

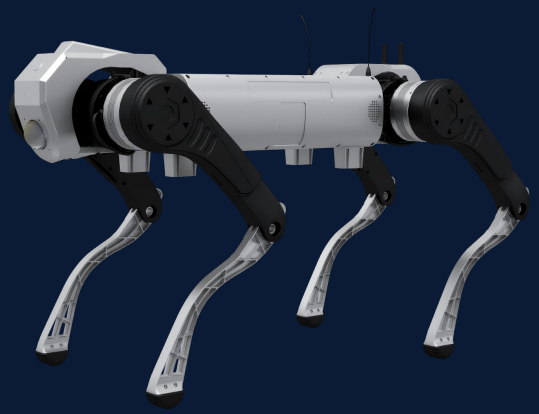
STEP 4 — **Auto-Annotation & Programmatic Augmentation:**

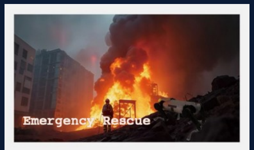
Integrates YOLO and SAM2 for object detection, plus HaMeR for hand pose and contact torque estimation. Supports lighting and viewpoint augmentation.

STEP 5 — **Standardized Packaging & Universal Export:**

One-click export to ROS Bag, PyTorch Dataset, and TFRecord- ready for mixed-model training out of the box.

Robo dogD1 MAX PRO





Highlights

Ultrahigh Payload

Supports sustained effective payload of 50kg, meeting the demands of heavyload scenarios such as integration with large payload modules, overweight transportation, and material handling.

Reliable Battery Life

Equipped with a 2160Wh highcapacity battery featuring exceptional power conversion efficiency. It delivers 2.5 hours of continuous operation under rated load, and up to 5.5 hours (industryleading) in idle conditions. The system also supports hotswappable batteries, with an optional autonomous charging dock available.

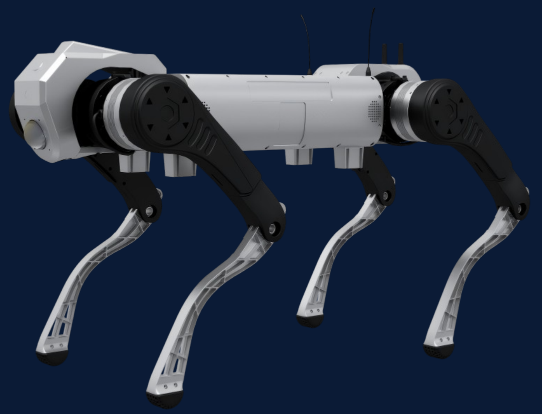
Outstanding Mobility Performance

Boasts exceptional agility and allterrain adaptability. Capable of smoothly climbing 30cm continuous staircases and open industrial stairs, with a robust selfrighting mechanism in place.

Superior Environmental Adaptability

With IP67 industrialgrade protection, it operates reliably across a wide temperature range of 20°C to 55°C and in water depths up to 20cm. Built to withstand extreme cold, scorching heat, wind, and rain, ensuring stable performance in harsh conditions.

Robo dogD1 MAX PRO



Functions

Standard Motions: Standing, Lying Down, Damping, Force Unloading; Stair Climbing (Up/Down), Obstacle Crossing
RealTime Image Transmission: Supported
Secondary Development: Supported
4G Communication: Optional

Performance Parameters

Continuous Walking Speed: 1.5m/s
Maximum Speed: 3m/s
Rated Payload: 50kg
Continuous Stair Climbing Height: 30cm
Maximum Climbing Slope: 45°

Robot Joint Parameters

Maximum Joint Torque: 180N·m
Joint Motion Range (Body): 45° ~ 45°
Joint Motion Range (Thigh): 200° ~ 20°
Joint Motion Range (Calf): 30° ~ 160°

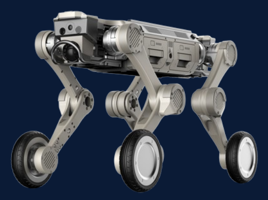
Parameters

Standing Dimensions (L × W × H): 1230mm × 530mm × 730mm
Prone Dimensions (L × W × H): 1313mm × 530mm × 300mm
Total Weight (Including Battery): 68kg
Operating Temperature: 20℃ ~ 55℃
Protection Class: IP67
Battery: Rated Capacity 45Ah, Voltage 48V
Charging Time: 4.5h (2h Fast Charging Optional)
Battery Runtime: 5.5h (No Load), 2.5h (Full Load)
Maximum Range (Full Load): 13.5km

Sensing Unit

LiDAR: 1 × 96channel LiDAR (Frontmounted), Coverage Angle: 360° × 90°, Coverage Diameter: 120m, Accuracy: 1.5cm
WideAngle Camera: 1 × WideAngle Camera at the Front and 1 at the Rear, Industrial Grade 2Megapixel, Field of View: DFOV 150°, HFOV 130°, VFOV 73°
IMU: Standard Configuration (SC)

Robo dogD1 MAX



Highlights

1. Core Performance Advantages
Engineered for industrialgrade performance
Low selfweight paired with high payload capacity
Longlasting battery life for extended operations
Robust environmental protection design
Agile yet stable locomotion for demanding tasks

2. Intelligent Locomotion & Terrain Adaptability

Reinforcement learningbased motion control model
Builtin capabilities: impact resistance, disturbance rejection
Fall selfrecovery mechanism for reliable field use
Dynamic adaptive control for complex environments
Autonomous adaptation to diverse terrains and road conditions

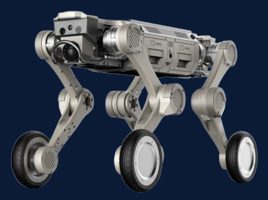
3. FullScene Perception & Navigation

Multimodal perception system: vision, LiDAR, ultrasonic, and INS
Environmental detection and mapping capabilities
Autonomous navigation in unstructured spaces
Realtime target recognition and tracking
Enables intelligent, collisionfree operations

4. MultiScenario Application Capabilities

Rich set of power and communication interfaces
Versatile for industrial and public safety use cases:
Security patrol & surveillance
Power grid inspection
Logistics and heavy payload transport
Onsite task execution & response
Personnel search & rescue (SAR)

Robo dogD1 MAX



Basic Information

Material: Aluminum Alloy + HighStrength Engineering Plastic
Standing Dimensions (L × W × H): 920mm × 480mm × 570mm
Prone Dimensions (L × W × H): 930mm × 630mm × 200mm
Total Weight (Including Battery): 40kg
Operating Temperature: 20℃ ~ 55℃
Protection Class: IP67
Battery: Rated Capacity 10Ah, Voltage 54V
Charging Time: ＜ 2h
Battery Runtime: 5h (No Load), 2.5h (Full Load)
Maximum Range: 29km (No Load), 18km (Full Load)

Performance Parameters

Maximum Speed: 8m/s
Rated Payload: 25 ~ 30kg
Continuous Stair Climbing Height: 22cm
Maximum Climbing Slope: 45°
Jumping Height Off the Ground: 80cm
Maximum Obstacle & Ditch Crossing Height: 80cm
Loaded Obstacle Crossing Height: 50cm
Minimum Passable Width: 50cm

Robot Joint Parameters

Maximum Joint Torque: 150N·m
Joint Motion Range (Body): 30° ~ 40°
Joint Motion Range (Thigh): 160° ~ 160°
Joint Motion Range (Calf): 160° ~ 160°

Sensing Unit

LiDAR: 1 × 192channel LiDAR at the front and 1 at the rear, with a coverage angle of 360° × 90° and a coverage diameter of 120m
Camera FOV: 1 × FPV camera at the front and 1 at the rear, industrialgrade 8megapixel, field of view: DFOV 122°, HFOV 111°, VFOV 70°
Fill Light: 4 × Fill Lights, 1 on each side of the FPV cameras
IMU: 1 × AutomotiveGrade IMU
RTK Module: 1 × GNSS RTK
Ultrasonic Sensor: 1 × Ultrasonic Sensor on each side, supporting obstacle recognition within 5m

Robo dog D1 Ultra

Highlights

Designed for specialized and industrialapplications

IndustrialGrade Compact Quadruped RobotDesigned for specialized and industrialapplications requiring reliability, mobility, and flexible deployment.

Tackle various complex terrains

Reinforcement learning (RL)based gait control enables autonomous adaptation to diverse and complex terrains, with builtin selfbalancing, antitipping, and disturbance rejection capabilities

Maximum running speed of 3.7 m/s, maximum climbing/descending slope angle ≥ 30°

Forward or vertical jump liftoff height up to 35 cm

Supports continuous climbing of stairs with a maximum riser height of 16 cm

Robot Joint Parameters

Maximum Joint Torque: 48N·m

Joint Motion Range (Body): 28° ~ 28°

Joint Motion Range (Thigh): 170° ~ 66°

Joint Motion Range (Calf): 35° ~ 156°

Sensors

IMU: Standard Configuration (SC)

Camera FOV: 1 Camera with 1920×1080p Resolution, Field of View: DFOV 122°, HFOV 111°, VFOV 70°

Performance Parameters

Maximum Speed: 3.7m/s

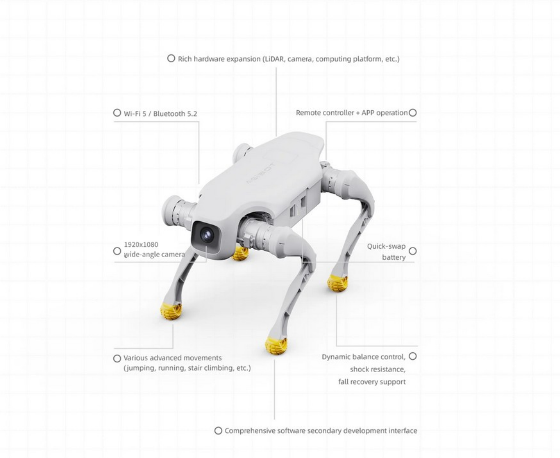
Rated Payload: 5kg

Continuous Stair Climbing Height: 16cm

Maximum Climbing Slope: 30° in Standard Environment, Up to 40° in Extreme Conditions

Jumping Height Off the Ground: 35cm

Robo dog D1 Pro



Robo dog D1 Pro

Designed for specialized and industrialapplications

IndustrialGrade Compact Quadruped RobotDesigned for specialized and industrialapplications requiring reliability, mobility, and flexible deployment.

Highlights

Powerful Locomotion to Tackle All Complex Terrains

Reinforcement learning (RL)-powered gait control enables autonomous adaptation to diverse and challenging terrains.

Built-in capabilities: self-balancing, anti-tipping, and disturbance rejection for stable performance in dynamic environments.

-Prone Dimensions (L × W × H): 675mm × 435mm × 145mm

-Total Weight (Including Battery): 15.5kg

-Operating Temperature: 0℃ ~ 40℃

-Protection Class: IP54

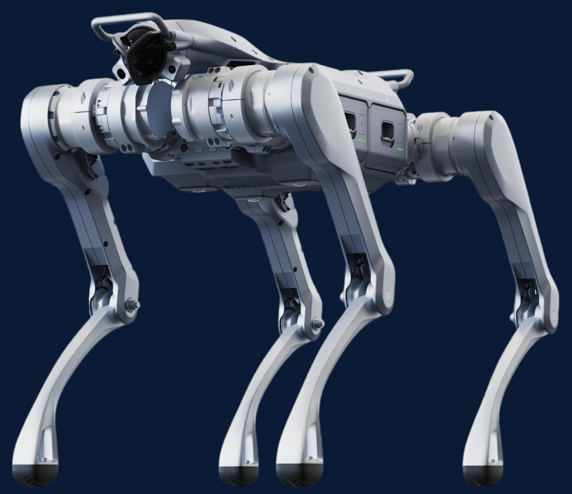
-Battery: Rated Capacity 4.6Ah, Voltage 43.2V

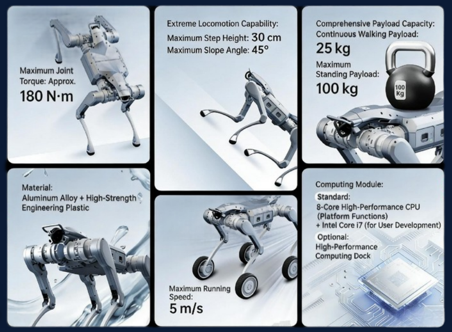
-Charging Time: 1h

-Battery Life: 1–2h

-Maximum Range: 6km

Robo dog





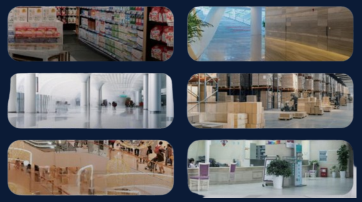
Functions

SLAM Localization and Mapping
Autonomous Navigation
Omnidirectional Obstacle Avoidance
Point Cloud Surround View
Intelligent Management and Scheduling System
Autonomous Charging
High reliability
Aftersales support

Parameters

Standing Dimensions (mm)
820 × 440 × 570
Weight
37 kg
Continuous Working Payload
25 kg
Max. Continuous Step Height
30 cm
Max. Step Height
80 cm
Limit Test Speed
5 m/s
Max. Operating Speed
3.7 m/s
Max. Climbing Angle
Approx. 45°
Trench Crossing Width
NoLoad Endurance Time / Range
5 h / 20 km
Effective Payload Endurance Time / Range
3 h / 12.5 km
Battery Capacity
907 Wh

Commercial & Industrial Cleaning



Applicable Floor Types

Functions

Allinone vacuuming & scrubbing
Laservision fusion positioning
Auto charge, water refill/drain, selfcleaning
Highefficiency edgetoedge cleaning
Low maintenance, long service life

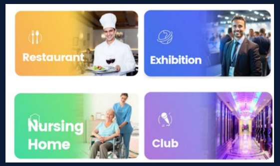
Application Scenarios

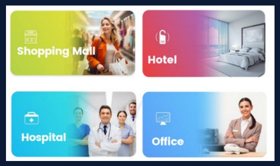
Large supermarkets
Office buildings
Transportation hubs
Factories
Commercial
Hospitals

Parameters

Garbage collection rate
95%
Overall noise
<68 db(A)
Machine weight
170 kg
Extra large volume of water tank
90 L
Selfcleaning dirty water tank
4 min
Floor scrubbing endurance
3 h
Battery specification
24V 80 Ah
Scrubbing pressure
25 kg
Maximum efficiency
1980 m^2/h
Scrubbing width
550 mm
Lifespan of Fan
10000 h
Water tank capacity
8 L
Water filling speed
710 L/min
Dirty water discharging speed
1015 L/min
Maximum power
1800 W
Rated input voltage
200240 V

GreetingBot Star Key Specifications





Size	558mm×525mm×1350mm
Net Weight	56kg
Screen Size	14-inch; 1080 HD
Chest Part Load Weight	5Kg
Narrowest Passage Width	70cm
Hardware Platform	Qualcomm 8-core chip + Industrial-grade MCU
Navigation Sensor	Li DAR*1 + Depth Vision Camera*3 + Fisheye Camera*2 + Infrared
Camera*2 + Odometer + IMU*1	Microphone Array
6-Mic arrays, 360° sound source positioning,	5 m sound range
Battery Life	12 H (typical cruising scene)
Charging Time	4.5 H
Supports Secondary Interface	USB 2.0 A port*2, Type-C port*1; LAN 100Mbps Ethernet port *1; DC
power port 12V 3A	Network Supports
4G (supports TDD-LTE, FDD-LTE), Wi Fi supports 2.4G/5G	

Workbot

Functions

Unified Integrated Control, No Interface Boundaries
SelfDeveloped Positioning Compensation, EnvironmentResilient
Intuitive HMI, OneClick Command Execution
Dual LiDAR + Dual Vision + Collision Detection, Enhanced Safety
Rapid Battery Swap, 24/7 Continuous Operation
Modular Chassis + Backpack Design, Maximized Flexibility

Application Scenarios

Automotive: LineSide Feeding, Flexible
CNC: Machine Tending, MultiMachine
3C: Component Handling, MultiSKU
Logistics: Tote/Pallet Handling, Automation
General Industry: Flexible Line Retrofit

Parameters

Dimensions (L×W×H)
950×630×850 mm (excluding arm)
Manipulator
Reach 1327 mm, payload 12 kg (incl. end effector)
Weight & Payload
Weight < 300 kg, usable payload 150 kg
Speed
Arm TCP ≤1.8 m/s, travel ≤1.2 m/s
Mobility
Obstacle ≤10 mm, gap ≤20 mm
Passage Width
Travel ≥850 mm, turning ≥1250 mm
Protection
Body splashproof, arm IP68
Accuracy
Station ±10 mm, comprehensive ±0.5 mm
Endurance
Single battery 3 h, dual batteries 6 h
Cycle Time
20 s
Radar Layout
Diagonal dual radar
Rotation Diameter
1000 mm

FMC^3 Brain+ODM

Seven-degree-of-freedom bionic arm

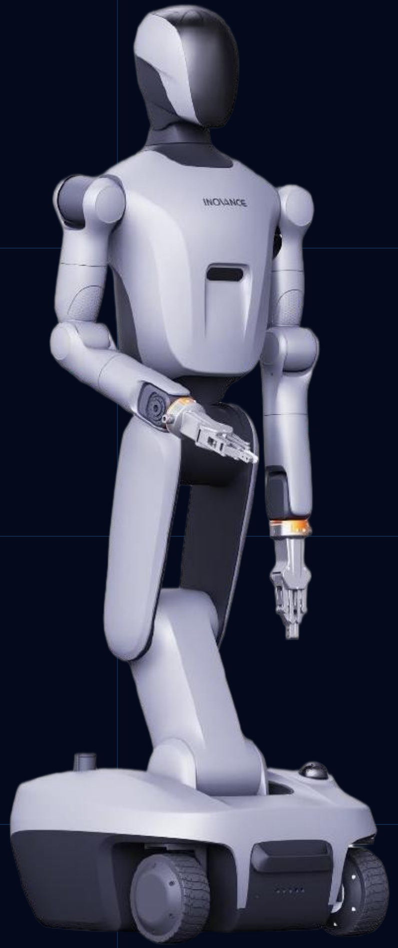
Dual-arm rated payload 7Kg, maximum 10Kg
Seven-axis anthropomorphic configuration, low weight-to-payload ratio
Seven torque sensors for compliant force control

Omnidirectional Mobile Chassis

Compact dimensions:594X591mm
Operating in narrow workstations and aisles
Omnidirectional movement, in-place rotation
Maximum Speed: 1.5m/s

Power management system

Endurance time: 8h
Supports hot-swap battery exchange



Embodied Intelligence

Supports high computing power of 2070 Tops
Compatible with mainstream Embodied Large Models
Compatible with multiple data acquisition devices

Quick-change End Effector

Supports quick-change for dexterous hands, grippers, and fixtures

Full-body Motion Control System

Full-body Coordinated High-frequency Motion Control
Supports ECAT hard master station, high reliability
Waist pitch and 180° rotation, for larger and more efficient working space

Navigation system

Multi-sensor fusion navigation: 3D LiDAR + 2D LiDAR + 3D camera

Application Scenario

Production Line Loading and Unloading

Logistics Sorting

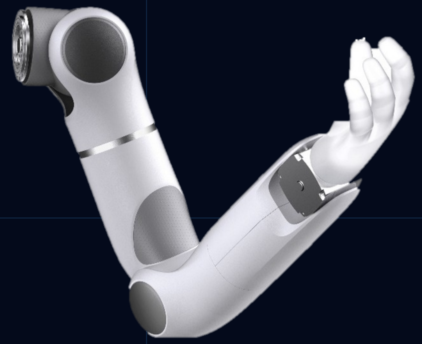
Unmanned Retail

Data Collection Training

Research and Education

Wellness and Elderly Care Collaboration

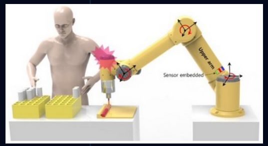
Humanoid upper limb, compliant control, safe human-robot collaboration



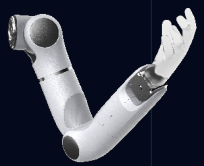
3.5 kg / 3.5 kg
Rated / Max. load

7.5 kg
Weight

Weight: 7.5 kg



Force-controlled insertion: Active force-position coupling for correction, dynamic adjustment of sub-millimeter deviations, ensuring successful completion of precision insertion tasks



Rated/Max. load: 3.5 kg/3.5 kg

01 **Humanoid configuration for confined space operation, reducing wrist space requirements by 50%+**

02 **Smooth motion control, stable grasping, maximum overshoot ≤ 1mm, settling time ≤ 0.5s**

03 **Seven-joint force sensing, high-precision force control,1000Hz full-body force control frequency**

04 Safe collaboration: < 1 N high-precision detection, stop within < 1s in case of collision, ensuring human safety

05 Vibration suppression: High-efficiency vibration damping, precise start-stop, precise compensation of joint flexibility, reduced vibration, no jitter during heavy-load operations

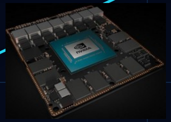
06 Suitable for operation within a 60 x 60 cm workspace, with minimal motion interference, compact wrist design for greater reachable positions

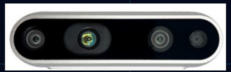
Refined and User-Friendly Infrastructure – Customer focus is on AI model deployment and research

Vision System Data

High Computational Power – NVIDIA Thor

Embodied AI Model





Interface Invocation

Motion Control System

Full-body Motion Control

Ethernet Interface
(Motion Control Message Interface)

Seven-Axis Bionic Arm

Folding Mechanism

Omni-directional Chassis

Motion Optimization and Coordinated Movement

- AI Hardware Platform with Underlying Motion Control Hardware Platform, Operating System, Message Interface, and SDK; comprehensive infrastructure configured in a single deployment
- The client AI system invokes perception data via the SDK, and the motion control interface sends all degrees of freedom space positions in one command for the robot to achieve full-body coordinated motion control

Cobot

Functions

Dualarm collaborative mobile manipulation
Omnidirectional movement + waist lift/pitch
Multimodal sensing & 1ms fast response
Quick deployment, outofbox use

Application Scenarios

Sorting
Handling
Loading & Unloading
Flexible Grasping

Parameters

Configuration
2×7axis arms, 4×vision cameras, 1×mobile chassis, 1×lifting unit
Total DOF
21
Waist Movement
Lift 0–800 mm; Pitch 0°–90°
Neck Movement
Rotate ±90°; Pitch 0°–45°
Arm Payload
5 kg
Dimensions & Weight
750×600×1565 mm; 200 kg
Controller
Intel i512th Gen, 128G SSD, WIFI6
Power & Interface
DC 30–60 V, 1500 W; EtherCAT, 4×GigE LAN
Environment
IP20, 0–50 °C
Material
Steel + Aluminum Alloy
Efficient
DualArm Collaboration
DualArm Coordination
& Full Workspace Reachability
MultiModal Perception
& Safety Assurance

Industrial Service Robot

Functions

Industrial
Service Robot
Perform highprecision assembly and manipulation
Move flexibly in all directions
Interact intelligently with humans
Execute realtime remote operation
Ensure safe operation in all scenarios
Work continuously with autonomous charging
Support fast application development

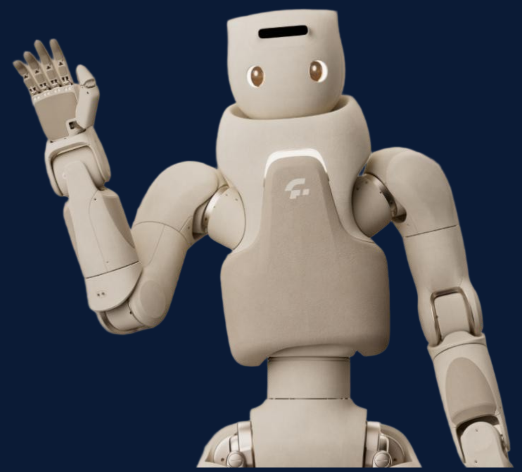
Application Scenarios

Logistics sorting
Highprecision assembly & operation
Industrial inspection
Guided tours
Security check
R&D support

Parameters

Body
26 DOF; humanlike motion; 5-DOF waist-leg system
Mobility
4wheel steering; omnidirectional movement; inplace rotation; crab walk; max speed 1.5 m/s
Arms
7DOF arms; ≤0.5N force control precision; submillimeter accuracy; 5 kg payload
Hands
Dexterous hand (optional); 3D tactile sensing; quick replacement; full joint torque sensing
Computing
NVIDIA Jetson Thor 5000 (2070 TFLOPS FP4);
Vision
Binocular stereo camera; RGB-D cameras; fisheye array; 360° surround view; 2× LiDAR
Audio
Mic array; sound source localization; 8W speaker
Power
Dual hot-swappable batteries; autonomous charging; 24/7 operation
Safety
360° perception; arm collision detection; active-passive protection
Neck
Roll, Pitch, Yaw
Waist & Legs
3× waist joints; 2× leg joints; 360° coverage
Ecosystem
Genie RL Toolkit; GDK; AGIBOT WORLD dataset; Genie Envisioner foundation model

Humanoid Home Service Robot



Functions

Whole-body teleoperation & human motion mapping
Precise manipulation with multi-DOF dexterous hands
Emotional interaction & social companionship

Application Scenarios

Social & emotional companionship
Public service & guidance
Elderly care & assistance
Rehabilitation & health support
Family accompanying & interaction
AI research & development
Assisted care, rehabilitation training support
24/7 operation with hot-swap batteries
Customizable functions

Parameters

Height
165 cm
Weight
71 kg
Total DOF
55
Hand
12-DOF dexterous hand
Arm Payload
3 kg per arm
Actuator
Self-developed highperformance integrated actuator
Battery
Dual hot-swappable
Battery Life
3 hours per battery
Operation
24/7 continuous operation
Vision
RGB + structured-light camera; facial recognition & tracking
Audio
4-mic array; echo cancellation; sound source localization
Tactile
Distributed pressure sensors for touch interaction
Control
Wholebody teleoperation; low-latency motion mapping
Structure
Modular design; compact & lightweight
Versions
GR-3: soft skin; GR-3C: reinforced shell + LED ring head

GreetingBot Nova Service Robot

Application Scenarios

•Showrooms & exhibitions
•Reception desks & office buildings
•Museums, galleries, guided tours
•Hotels, hospitals, retail stores

Functions

•Large language model voice interaction in 30+ languages
•Smart reception, intelligent Q&A, guidance and tour explanation
•Dynamic promotion with 14-inch FHD customer-facing screen
•Marker-free deployment, autonomous navigation and obstacle avoidance
•Automatic recharge for continuous front-of-house service

Parameters

Size
558 × 525 × 1350 mm
Net Weight
56 kg
Screen
14-inch; 1080 FHD
Camera
13M high-definition head camera
Mobility
0.5-1.2 m/s adjustable cruising speed
Endurance
12-14 h typical cruising scene
Charging
4.5 h via automatic charging dock or cable
Interfaces
USB 2.0, Type-C, LAN, DC power 12V 3A
Business Value
Reduce routine reception work; improve visitor engagement

IBen-A03 Service Robot

Application Scenarios

•Banks & government service halls
•Exhibition halls and showrooms
•Reception, greeting and consultation
•Guided tours, entertainment interaction

Functions

•Humanoid design with 10-DOF flexible dual arms
•Reception & greeting with user recognition and custom greetings
•Business consultation via business knowledge base and LLM connection
•Indoor autonomous navigation, leading and real-time obstacle avoidance
•Android SDK / web adaptation for extend development

Parameters

Dimensions
54 × 54 × 155 cm
Weight
79 kg
Arms
5-DOF per arm; shoulder, elbow and finger movement
Screen
15-inch HD touchscreen
Sensors
30 m radar, 3 ultrasonic sensors, 1 depth camera
Compute
RK3588 8-core CPU; 8G+64G
Battery
30Ah LiFePO4; 8 h usage, 6 h charge
Speed
0.3-0.7 m/s adjustable; ±5 cm positioning accuracy

FMC^3 Robotics

Thanks

www.fmc3robotics.ai